

In an operating environment shaped by predictably unpredictable geopolitical shocks, timely intelligence becomes a cost lever as well as a safety lever.

Our [Strategy team](#) have put together a topical business case to provide insights into the importance of geopolitical risk intelligence. Explore the insights below.



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Airlines' first order annual cost exposure

We estimate that the cumulative effects of airspace restrictions, operational disruption, and geopolitically driven volatility cost the global airline industry in the order of USD 5 – 12 billion annually.

This represents a small proportion of total industry revenues but a disproportionately large drag on profitability, reflecting the sector's thin margins.

We estimate that 20-30% of these costs may be avoidable through the adoption of geopolitical risk-intelligence tools that provide timely, predictive insight.

This estimate is based on an active global commercial fleet of approximately 31,000 aircraft, comprising around 25,000 narrow-body and 6,000 widebody aircraft. The breadth of the estimated range reflects the number of underlying variables and the resulting year-to-year variability in geopolitical developments.

To address this uncertainty, 10,000 simulations were conducted, with USD 5 billion representing the 25th percentile and USD 12 billion representing the 75th percentile of outcomes.

This USD 5 – 12 billion estimate is by nature conservative: it captures direct, first order operational disruption (e.g., rerouting and delays), but does not quantify broader knock-on effects - for example, when an aircraft is diverted or grounded and cannot operate its

subsequent scheduled flights, as well as the resulting rerouting costs for passengers and crew.

In practice, geopolitical events can ripple through the global economic system and affect airlines well beyond those operating in and around the immediate conflict zone.

For example, a regional war can tighten jet fuel supply and raise prices for airlines globally. A scenario where this type of event causes a USD 50 per barrel increase in jet fuel, sustained for ~3 months, implies ~USD 30bn of incremental aviation industry fuel cost if purchased at spot.

Even allowing for hedging, such an event could still disrupt global supply chains for essential aircraft components, including semiconductors and avionics.

These second and third-order impacts are difficult to measure reliably at an industry level. However, it is reasonable to expect that the exposure could greatly exceed the more readily quantifiable first-order impacts.

Getting ahead of government intel

Reactive approaches to geopolitical risk are increasingly insufficient.

Global air travel is expanding rapidly, with around 200,000 flights operating daily and this figure projected to roughly double by 2042. At the same time, geopolitical tensions and airspace restrictions are forcing more flights into fewer corridors, creating unprecedented operational challenges.

Risk is no longer confined to traditional conflict zones; disruptions originating in one region can quickly ripple across global networks. Rising traffic, shifting conflict dynamics, and constrained airspace are collectively increasing both risk exposure and operational costs for airlines.

Aviation leaders, including heads of security, operations, and insurers, emphasise that rising global instability is forcing airlines to shift from reactive approaches to proactive, real-time geopolitical monitoring.

For most of aviation history, airlines mainly relied on government-led intelligence because it was the only authoritative, structured source of information about geopolitical threats, airspace restrictions, and security risks.

Regulators, embassies, and national security agencies controlled access to updates on conflict zones or safety concerns, and airlines depended on these official channels to know where it was safe to fly.

This system made sense in an era when governments held the best data; but it also meant airlines often received information that was slow, limited, or difficult to share across their operational teams.

Limitations of government-led geopolitical monitoring

Limitation	Illustrative industry feedback	
Biased	Long-haul airline, Middle East	<i>"Publications from the authorities can be filtered through political lenses, which means risks communicated can be downplayed or overplayed."</i>
Slow	Business aviation operator, Europe	<i>"Regulators announce closure of airspace, but they are often lagging what geopolitical events are unfolding on the ground."</i>
Generic	Oil and gas company, Global	<i>"Often the reports being published by government agencies are for a wide audience and too general to be operationally useful in the aviation context."</i>
Not easily distributable internally	Leisure airline group, Europe	<i>"Government intelligence, whilst very good, is inherently not shareable with a wide community in our organisation."</i>

Costs of disruption

The costs of disruption caused by geopolitics are multi-fold.

Geopolitical disruption is costly because it impacts multiple P&L lines simultaneously, through factors such as fuel burn, crew duty constraints, missed airport slots, diversions, fines, and insurance costs.

Cost Type	Description	Quote
Direct operational impacts	Rerouting	Longer routings to avoid unsafe, or perceived unsafe, airspace increase fuel burn, flight time, and ANSP charges, making some routes unfeasible due to aircraft range limits or nonviable due to higher operating costs <i>"Rerouting around Iranian airspace during pre-2026 regional tensions added 45–50 minutes of flying time for Far-East routes costing us more than £1.5 million over a season."</i> Leisure airline group, Europe Note: These reroutings highlight how precautionary (and in some cases arguably premature) decisions, rather than imminent threats, can nonetheless result in significant cost exposure.
	Diversions	Unexpected diversions and emergency repositioning create operational costs. <i>"When we took the call to suspend our operations in Israel when the Gaza War began, and had to reposition aircraft and crew, it was a major and costly network decision."</i> Global freight carrier
	Route suspension	Route suspensions in higher-risk operating environments lead to lost revenue. Furthermore, aircraft and crews stranded at the start of a crisis create immediate operational and recovery costs, while delayed restarts after a disruption further reduce passenger and cargo revenue even after airspace risk has returned to acceptable levels, as operators wait for competing airlines to make the first move. <i>"There is a scenario where airlines suspend flights but know they will eventually restart them. When that restart is delayed, for example, starting the route one or two months later, the airline incurs an opportunity cost equal to those one or two months of lost revenue."</i> Geopolitical risk intelligence vendor
Indirect operational impacts	Asset exposure	Aircraft face increased risk of missile, drone, or conflict-related damage. <i>"We operate in regions with missile threats and no-warning weapons testing; it's a volatile environment for aviation."</i> Global air-cargo operator
	Crew exposure	Crew and ground staff may require rerouting, relocation, rapid evacuation or repatriation, creating knock-on rostering impacts when staff are out of position <i>"When local security deteriorates, we become concerned about getting our people out safely and ensuring they are not stuck in the wrong place at the wrong time."</i> Long-haul carrier, UK
Overhead costs	Insurance premiums	Higher insurance premiums and restrictions when flying into risky regions. <i>"Flights into high-risk areas require higher premiums and restrictions, such as no overnight stays or refuelling."</i> Speciality insurer underwriter
	Risk staffing	More analyst hours required to monitor conflicts and assess risks. <i>"For an organisation of our scale, conducting route risk assessments manually would demand a dedicated team of 10 to 12 analysts."</i> Long-haul carrier, UK

Moving proactively

Commercial geopolitical risk intel solutions enable airlines to move proactively - rather than reactively.

Due to the shortcomings of relying on slow, inconsistent government intelligence, and the heavy operational burden of manually interpreting raw information for aviation, airlines increasingly look to commercial risk-intelligence platforms.

Best-in-class risk intelligence means airlines get fast, verified, aviation-specific insights about any threat that could affect a flight, crew or airport. It goes far beyond news scraping or waiting for late government notices. Instead, solutions exist that integrate data from multiple sources, including news platforms, social media, government notices, satellite and thermal imagery, aircraft movement data and filters it into accurate, unbiased, real-time and even predictive alerts.

By crosschecking data inputs and filtering out unreliable data, these platforms help ensure that risk assessments rely on objective and verified information rather than isolated or subjective observations. Airlines value this because decisions on overflight, routing, and airport safety and security must often be made within minutes, not hours, and are further strengthened by predictive alerts that enable contingency planning.

It also means the intelligence is interpreted, not just delivered raw. Leading platforms show what threats mean for actual flight paths, altitudes, aircraft types and crew safety. Historical trends, predictive analysis and easy-to-use dashboards help airlines assess whether risks are rising, falling or stable.

Ultimately, best-in-class intelligence supports safe but optimised operations; avoiding unnecessary costs while avoiding genuine hazards. It enables airlines to implement timely mitigation measures as risk rises - neither too early nor too late - while also supporting dynamic route reopening once risk returns to acceptable levels, with decisions clearly justified across the organization, including to crew and passengers.

Industry adoption remains limited, with most airlines operating below best practice.

Only an estimated 15–20% of airlines, typically larger and more safety-mature carriers, have adopted truly best-in-class, daily risk-intelligence tools. These airlines treat risk intelligence as essential infrastructure rather than a “nice to have,” and expect immediate alerts, deep analysis, forecasting, and aviation-specific insight.

By contrast, the majority of airlines continue to rely on government advisories, media reporting, manual checks, or generic intelligence tools not designed for aviation. Even among larger carriers, capability is often fragmented or incomplete. While adoption is accelerating in response to recent global events, the industry overall remains in the early-to-middle stages of maturity.



During the escalation in Myanmar's civil war, when some of our establishment partners were advising us to avoid the airspace entirely, our risk intel tool gave us confidence to continue operating safely at defined minimum altitudes.

Security Manager

Long-haul cargo airline



We spot upcoming peaks in military activity and adjust routes or pause flights early; keeping us operating longer and resuming faster than if we'd relied on government intel.

Security Manager

European carrier

Getting a real-time risk picture

The market is somewhat under-served, with few vendors capable of offering a comprehensive, real-time risk picture.

Capability comparison

Figure: Capability comparison of aviation intelligence and risk-monitoring providers, ordered by completeness of coverage on the evaluation categories.

Provider	Analyst verified aviation intelligence	Proactive alerts (push, not pull)	Quantified tactical risk forecasts with timings	Location-specific dynamic risk scores	Scalable (batch) flight risk assessment	Crisis response
Osprey Flight Solutions	✓	✓	✓	✓	✓	✓
Dyami	✓	✓	✓	✓	✗	✓
MedAire	✓	✓	✗	✓	✓	✓
DragonFly	✓	✓	✓	✗	✗	✗
Seerist	✗	✓	✓	✓	✗	✓
Control Risks	✓	✗	✗	✗	✗	✓
Ops Group	✓	✓	✗	✗	✗	✗
Horizon Intelligence	✓	✓	✗	✗	✗	✗

Justifying investment

 Has capability
  Has partial capability
  Doesn't have capability

Perceived intangibility of ROI is an adoption hurdle for some.

Many airlines struggle to justify investment because the financial benefits of risk-intelligence tools are not easily expressed in a simple, measurable way.

Many airlines we've spoken to say a major barrier to adopting commercial risk-intelligence solutions is that the ROI feels difficult to articulate or quantify.

While the operational value is often clear to them – better decisions, faster intelligence, and improved safety and security – the financial return is harder to convert into a simple number. Much of the benefit comes from “losses avoided”, which procurement teams struggle to treat as tangible.

Combined with tight budgets and competing priorities, this leads many airlines to continue relying on slower government intelligence or manual assessment processes rather than investing in commercial platforms.

This challenge is reinforced by organisational dynamics. Security and operational teams are, for valid reasons, not commercially driven, which can make it difficult to build a strong business case.

In hindsight, better intelligence would have changed some past decisions and avoided costly outcomes, but acknowledging this can be difficult, particularly where it implies that earlier judgments had material financial consequences.

As a result, there is often a tendency to protect the status quo. In these cases, airlines may rely on external justification, such as alignment with government guidance or peer decisions, rather than taking ownership of different, independent judgements enabled by next-generation tools.

While long-haul and high-exposure airlines see large, measurable returns, even airlines operating in supposedly low-risk regions are recognising that unexpected events can quickly turn risk intelligence into a critical operational asset.

Airlines also highlight that ROI varies dramatically depending on route network and exposure profile. Global carriers flying long-haul over conflict-affected regions see very large, measurable gains: airspace reopening, fuel savings, avoided detours, and reduced crew-time impacts. Some report that a single routing decision can outweigh the annual platform cost.



There's a large proportion of airlines that haven't sourced a geopolitical risk solution simply because it isn't a regulatory requirement, rather than because they've evaluated the economic case.

Former regulator

Many airlines operating in what they believe are stable environments don't immediately see the benefit; yet that stability is increasingly fragile.

Recent events across Europe show how quickly conditions can change, whether through geopolitical shocks or sudden, highly disruptive drone incursions.

In late 2025, for example, drone sightings shut down airports across Belgium, Denmark, Germany, and the Netherlands, including a complete suspension of flights at Eindhoven Airport, showing that even 'low-risk' regions can face major operational disruption with no warning.

Even adopters acknowledge that ROI fluctuates widely based on routes, exposure, disruptions, and seasonality, leaving non-adopters unsure how to quantify value.

Many airlines using these platforms generally describe payback in weeks rather than months, with most recovering the investment well within a single operational season. Others view payback as a longer-term accumulation of safety, efficiency, and compliance benefits. In both cases, many airlines are still making ROI judgments based on gut feel and intuition.

For those that already use these tools, they have become core infrastructure, making financial justification far easier than in the past. However, for airlines yet to adopt them, the perceived low measurability of returns remains a key reason for hesitation.

Getting ROI with the right solution

Scenario modelling shows that adopters of these solutions can expect a positive ROI.

A probabilistic approach to understanding ROI is needed because the disruptions driven by geopolitical tensions are highly variable by nature.

Because ROI is highly variable across fleets, routes, aircraft mix, and geopolitical exposure, many airlines struggle to form a confident financial case for adopting risk-intelligence tools. To address this uncertainty, we've developed a structured way to quantify ROI using a Monte Carlo simulation.

This approach reflects the reality that disruption costs depend on wide-ranging factors: aircraft types, daily flight volumes, percentage of routes exposed to geopolitical tension, the likelihood of diversions, typical delay minutes, and the rare but costly grounding of aircraft.

Monte Carlo is well-suited here because it helps model situations where uncertainty genuinely matters and simple averages hide meaningful variation.

In practice, the model runs ten-thousand simulations for an archetypal 100-aircraft airline, randomising inputs such as fleet mix, daily flight volume, disruption rates, diversion probabilities, additional flying minutes and the occasional multi-day grounding of aircraft.

Each run produces a different annual disruption cost, building a realistic distribution of low, median and high financial outcomes. A conservative mitigation rate (e.g., 10-20%) is then applied to estimate the portion of these costs that risk-intelligence tools could reasonably avoid.

The analysis also incorporates the per-aircraft insurance discount realised by adopting a geopolitical risk management solution.

Comparing those savings to the total solution cost (e.g., ~USD 10-12k per tail) produces a range of plausible ROI results, highlighting typical, downside and upside scenarios.

Quick payback

A typical airline can expect to recover the cost of a one-year subscription to a geopolitical risk-intelligence solution in around 6–8 weeks.

We modelled 10,000 scenarios to understand the payback period for adopting a geopolitical risk-intelligence solution.

Each simulation varied core drivers such as fleet mix, flight volumes, disruption rates, diversion probabilities and grounding events, producing a wide distribution of possible financial outcomes.

The results show that while payback periods naturally differ across operating profiles, the distribution clusters tightly enough to give a clear picture of the likely return.

Across all 10,000 simulations, the median scenario achieved payback in around 7 weeks, reflecting the cumulative effect of avoided disruption, shorter routings and faster operational decisions.

Early-benefit cases demonstrate even faster recovery, with 25% of scenarios paying back within 5 weeks.

Even under more conservative conditions; where disruption is less frequent, rerouting benefits are smaller, and grounding events are rare; 95% of cases reached payback within 24 weeks.

This indicates that, although individual airlines will experience different levels of impact, the underlying economics remain robust across a broad range of real-world conditions.

Modelled payback period for a geopolitical risk solution, 10,000 scenarios

Real-world benefits

Typical scenarios have been modelled, but real-world benefits can be even larger.

A. Black Swan events	▼
B. Opportunity-cost gains and second-order disruption impacts	▼
C. Small-fleet effects	▼

What airlines should do next

In an operating environment shaped by predictably unpredictable geopolitical shocks, timely intelligence becomes a cost lever as well as a safety lever

Priority actions for airlines:

Daily geopolitical risk intelligence

Airlines should integrate geopolitical risk intelligence into their day-to-day operational decision-making, rather than treating it as an occasional reporting exercise.

! **Geopolitical risk management as priority**

The C-suite should set a clear organisational expectation that proactive geopolitical risk management is a strategic priority, ensuring that investment decisions, governance structures, and cross-functional accountability all support the consistent use of high-quality intelligence in safeguarding the airline's network, financial performance, and long-term resilience.

Tools and scenarios

Airlines should present the business case for adopting risk-intelligence tools by using a range of realistic scenarios that capture how disruption varies in practice, rather than relying on a single financial estimate.

KPMG's independent assessment of the value of geopolitical risk intelligence is informed by multiple sources, with thanks to Osprey Flight Solutions for specific data contributions and support with production costs.



Flying Smarter - The business case for geopolitical risk intelligence

(PDF, 1.5MB)

 **Download the full report**

Get in touch

Implementing geopolitical risk intelligence solutions can deliver rapid payback for airlines, with a typical payback period of 6-8 weeks.

Connect with us today to explore how our strategic services can support your organisation on this journey to achieve results and cost-efficiency.

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Explore the business case for geopolitical risk intelligence

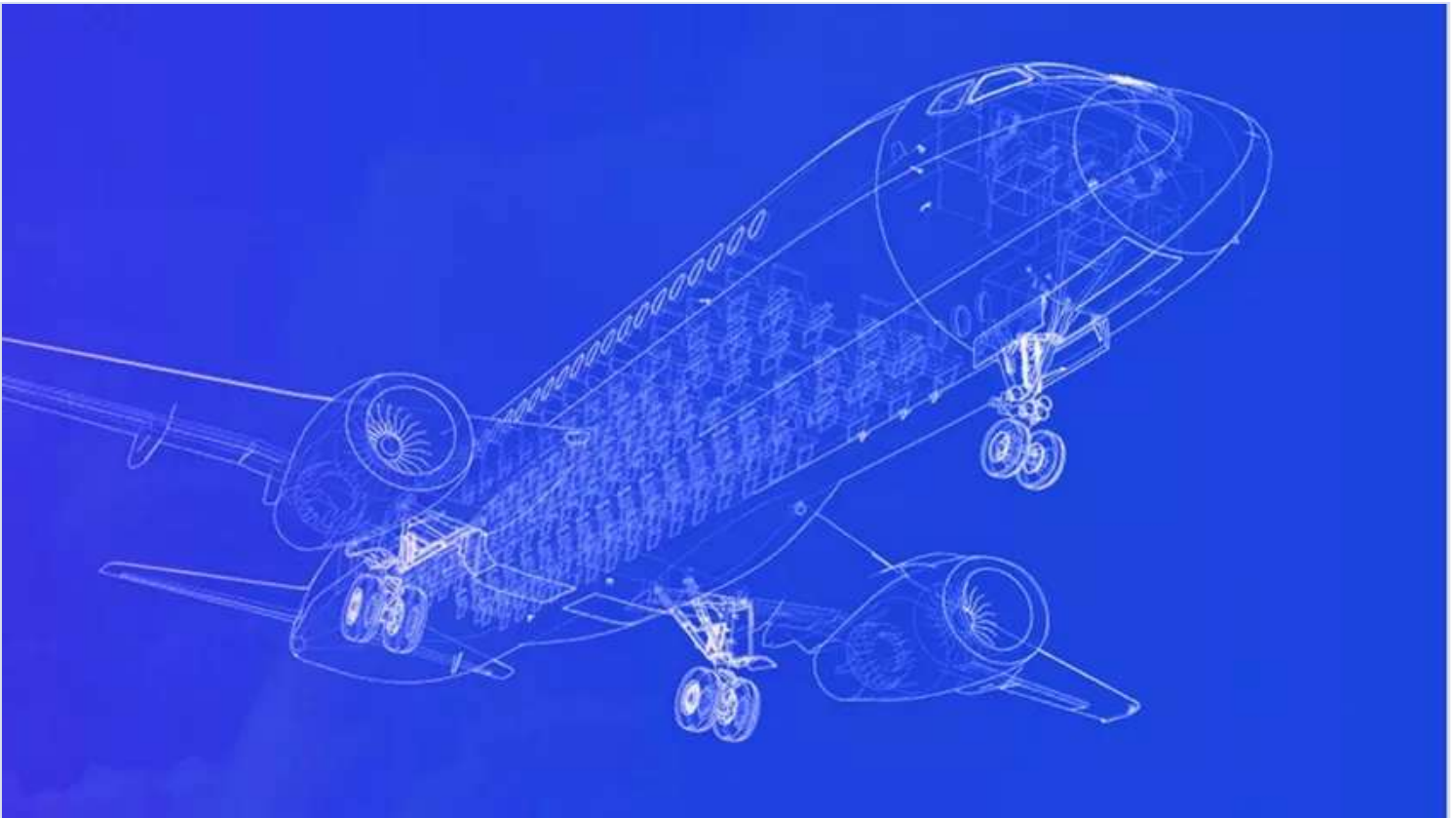
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Maintenance, repair and overhaul

As demand continues to outpace capacity, MRO is shifting from cost line to value lever across the global aviation ecosystem.

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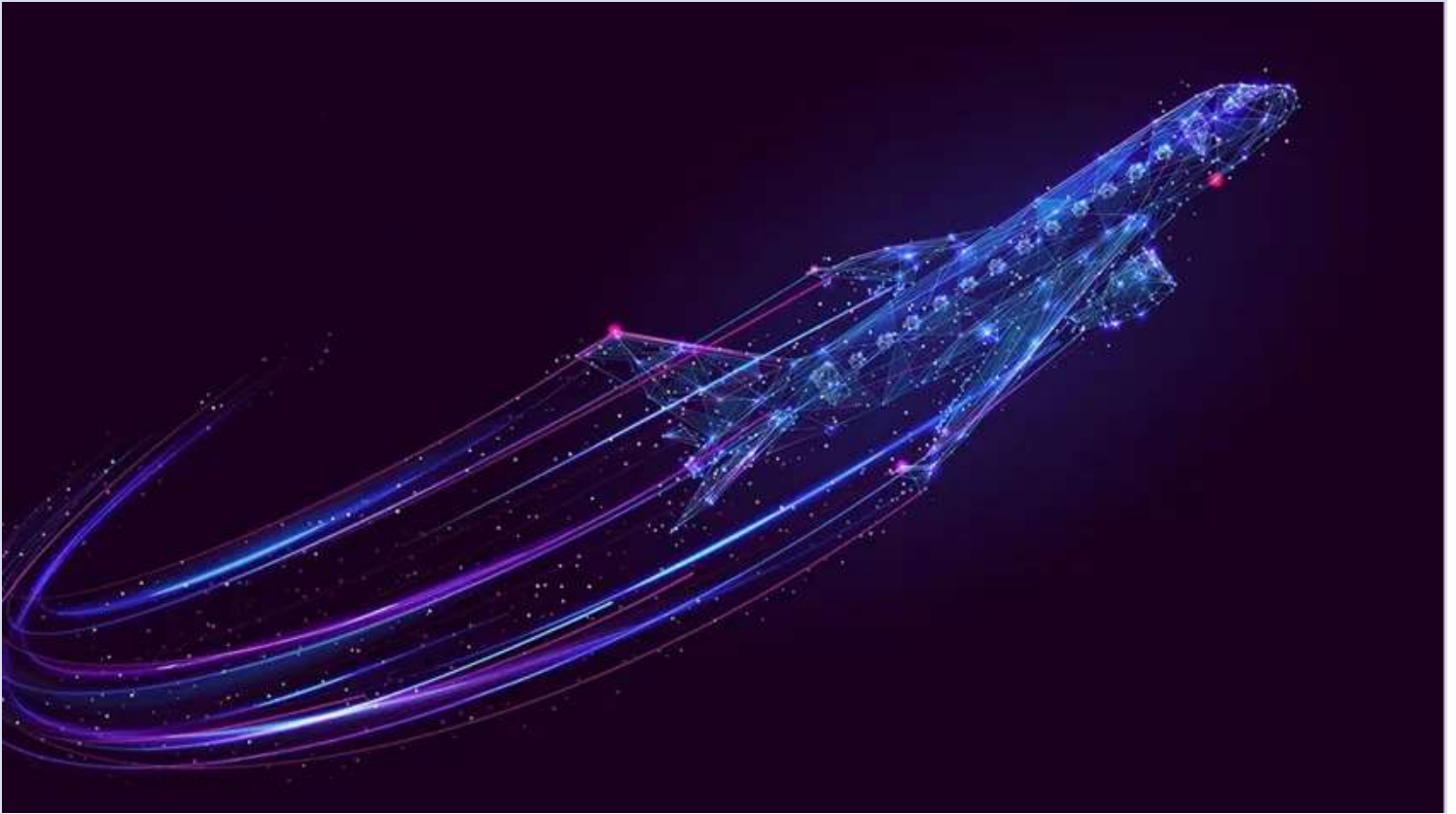
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